

PREDICTION: LEGITIMATE

RISK SCORE: 58 / 100 (MEDIUM RISK)

Scam Probability: 40.10% | Model Confidence: 59.90%

Banking Action: OTP VERIFICATION

Official Recommendation: OTP Verification Required. Moderate risk parameters detected. Challenge customer with SMS OTP or biometric 2FA authentication.

1. Transaction Details & Actual Parameter Values

|                  |                  |                          |             |
|------------------|------------------|--------------------------|-------------|
| Transaction ID:  | TXN-911606       | Transaction Amount:      | \$15,000.00 |
| Customer ID:     | C123456789       | Sender Old Balance:      | \$20,000.00 |
| Beneficiary ID:  | M987654321       | Sender New Balance:      | \$5,000.00  |
| Payment Type:    | TRANSFER         | Receiver Old Balance:    | \$500.00    |
| Device Type:     | Unknown Proxy    | Receiver New Balance:    | \$15,500.00 |
| Location Region: | High Risk Region | 6h Transaction Velocity: | 6 txns      |
| New Beneficiary: | Yes (New)        | Prior Fraud History:     | No          |

2. Composite Risk Score Breakdown & Latency Timing

| Risk Dimension             | Sub-Score | Status & Impact              |
|----------------------------|-----------|------------------------------|
| Model Fraud Probability    | 40.10%    | Weight: 70% (Score: 40.1)    |
| Amount Risk                | 20.0 pts  | Transaction size evaluation  |
| Velocity Risk              | 20.0 pts  | 6-hour transaction frequency |
| Device Risk                | 25.0 pts  | Device profile verification  |
| Location Risk              | 25.0 pts  | Geographic regional risk     |
| Beneficiary Risk           | 15.0 pts  | New beneficiary check        |
| Historical Behaviour       | 0.0 pts   | Prior fraud flag check       |
| Balance Behaviour          | 0.0 pts   | Account wipeout detection    |
| FINAL COMPOSITE RISK SCORE | 58 / 100  | MEDIUM RISK CATEGORY         |

|                          |                        |                         |                          |                          |                      |
|--------------------------|------------------------|-------------------------|--------------------------|--------------------------|----------------------|
| Feature Eng:<br>46.40 ms | Prediction: 1.59<br>ms | SHAP Engine:<br>2.11 ms | LIME Engine:<br>89.80 ms | Rendering:<br>1162.17 ms | TOTAL: 1302.80<br>ms |
|--------------------------|------------------------|-------------------------|--------------------------|--------------------------|----------------------|

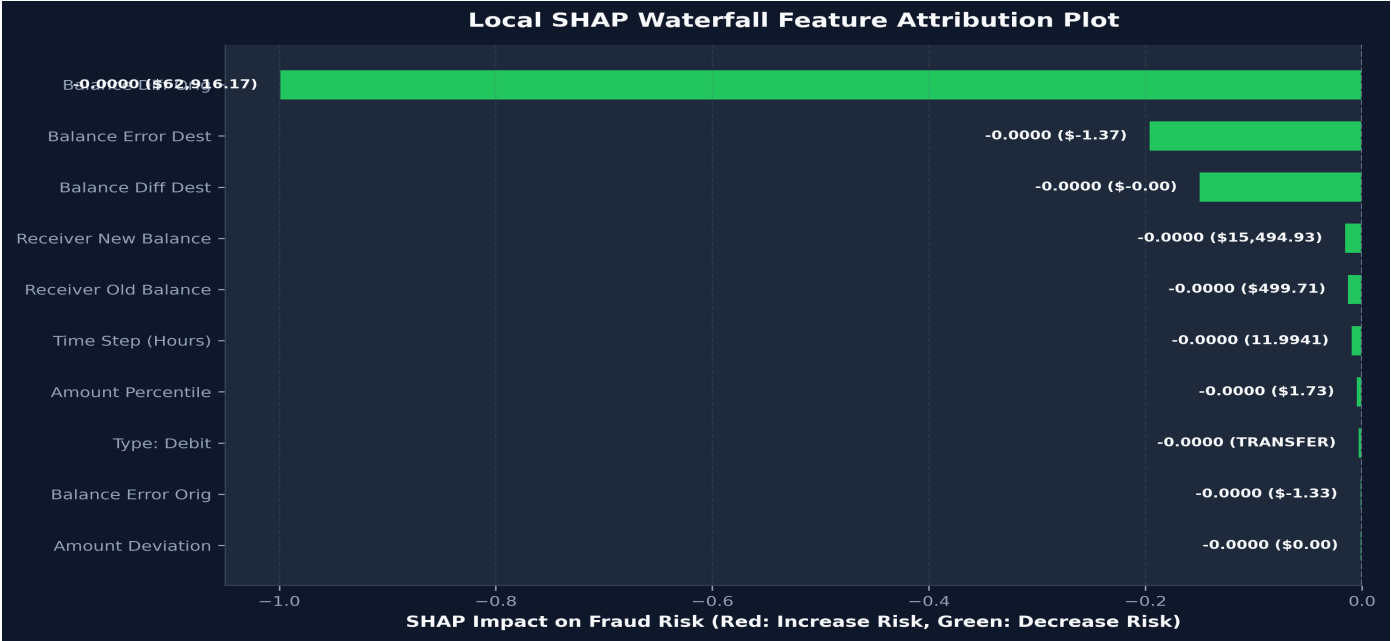
3. Banking-Grade Analyst Narrative Report

|                      |                                                                                                                                                              |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Transaction Summary: | Transaction TXN-911606 involving Customer C123456789 transferring \$15,000.00 via TRANSFER to Beneficiary M987654321 on Unknown Proxy from High Risk Region. |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                       |                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Behaviour Analysis:   | Sender account retained \$5,000.00 post-transaction. High velocity anomaly: 6 transactions executed within the last 6 hours. Device anomaly: Payment initiated from unverified device profile ('Unknown Proxy'). Geographic anomaly: Payment originated from high-risk or foreign region ('High Risk Region').                                  |
| High Risk Indicators: | Elevated Transaction Amount: \$15,000.00 exceeds single transfer threshold.; Rapid Transaction Spike: 6 transfers within a 6-hour window.; Unverified Device Profile: Initiated via Unknown Proxy.; Suspicious Geographic Location: Access from High Risk Region.; Unestablished Beneficiary: No prior transfer history with recipient account. |
| Safe Indicators:      | Account Balance Preserved: Remaining sender balance is \$5,000.00.                                                                                                                                                                                                                                                                              |
| Model Explanation:    | GuidedGuard Gradient Boosting model evaluated this payment pattern with a scam probability of 40.10% and classified it as 'Legitimate'. The composite banking risk score is categorized under MEDIUM RISK.                                                                                                                                      |
| SHAP Summary:         | Feature contributions align with baseline legitimate profile.                                                                                                                                                                                                                                                                                   |
| Final Recommendation: | <b>OTP Verification Required: Moderate risk parameters detected. Challenge customer with SMS OTP or biometric 2FA authentication.</b>                                                                                                                                                                                                           |

4. Top 10 SHAP Feature Contribution Attributions

| Feature Name         | Actual Value | SHAP Value | Contrib % | Direction      | Impact Level |
|----------------------|--------------|------------|-----------|----------------|--------------|
| Balance Diff Orig    | \$62,916.17  | -0.0000    | 45.47%    | Decreases Risk | High         |
| Balance Error Dest   | \$-1.37      | -0.0000    | 8.95%     | Decreases Risk | Medium       |
| Balance Diff Dest    | \$-0.00      | -0.0000    | 6.86%     | Decreases Risk | Medium       |
| Receiver New Balance | \$15,494.93  | -0.0000    | 0.75%     | Decreases Risk | Low          |
| Receiver Old Balance | \$499.71     | -0.0000    | 0.62%     | Decreases Risk | Low          |
| Time Step (Hours)    | 11.9941      | -0.0000    | 0.48%     | Decreases Risk | Low          |
| Amount Percentile    | \$1.73       | -0.0000    | 0.25%     | Decreases Risk | Low          |
| Type: Debit          | TRANSFER     | -0.0000    | 0.19%     | Decreases Risk | Low          |
| Balance Error Orig   | \$-1.33      | -0.0000    | 0.11%     | Decreases Risk | Low          |
| Amount Deviation     | \$0.00       | -0.0000    | 0.10%     | Decreases Risk | Low          |



5. Local LIME Decision Rules & Explanations

| Decision Rule Condition            | Rule Weight | Effect Direction | Domain Explanation                                                                                    |
|------------------------------------|-------------|------------------|-------------------------------------------------------------------------------------------------------|
| Sender Old Balance = \$20,035.35   | +2404.1819  | Increases Risk   | Feature condition 'Sender Old Balance = \$20,035.35' contributed positively to scam classification.   |
| Receiver New Balance = \$15,494.93 | +1859.3316  | Increases Risk   | Feature condition 'Receiver New Balance = \$15,494.93' contributed positively to scam classification. |
| Transaction Amount = \$14,970.39   | +1796.3870  | Increases Risk   | Transaction amount is significantly elevated relative to normal threshold.                            |
| Sender New Balance = \$5,011.35    | +601.3024   | Increases Risk   | Feature condition 'Sender New Balance = \$5,011.35' contributed positively to scam classification.    |
| Receiver Old Balance = \$499.71    | +59.9055    | Increases Risk   | Feature condition 'Receiver Old Balance = \$499.71' contributed positively to scam classification.    |
| Time Step (Hours) = 11.9941        | +1.3793     | Increases Risk   | Feature condition 'Time Step (Hours) = 11.9941' contributed positively to scam classification.        |
| Type: Payment = TRANSFER           | -0.0602     | Decreases Risk   | Payment type associated with routine account transfers.                                               |
| Type: Cash Out = TRANSFER          | -0.0597     | Decreases Risk   | Payment type associated with routine account transfers.                                               |
| Type: Transfer = TRANSFER          | +0.0597     | Increases Risk   | Transfer/Cash Out payment type has higher statistical fraud incidence.                                |
| Type: Debit = TRANSFER             | -0.0595     | Decreases Risk   | Payment type associated with routine account transfers.                                               |

Local LIME Surrogate Decision Rules Explanation

